July 17, 2023

Lab Tomorrow

Written HW 3 : Due tomorrow by midnight

Studio Thursday : Meet from Noon-2pm

Quizzes 1&2 : Friday!

Reflection 4 : Due Friday by midnight

Conceptual HW 4 : Due Sunday by midnight

Office Hours

Mondays	Noon-2pm
Thursdays	2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 3	**July 10** V2 CH 6	**July 11** Homework 2 Lab W3 - Using	**July 12** V2 CH 7	**July 13** V2 CH 7 Studio : Gauss Law	**July 14** V2 CH 7 Reflection
Week 4	**July 17** V2 CH 8	**July 18** Homework 3 Lab W4 - Electric Potential	**July 19** V2 CH 8	**July 20** V2 CH 9 Studio : V and C	**July 21** Reflection Quiz 1&2: CH 5-7
Week 5	**July 24** V2 CH 9	**July 25** Homework 4 Lab W5 - Potential & Field	**July 26** V2 CH 9	**July 27** V2 CH 10 Studio : Ohm's Law	**July 28** V2 CH 10 Reflection

Chapter 8

Capacitors

Capacitors

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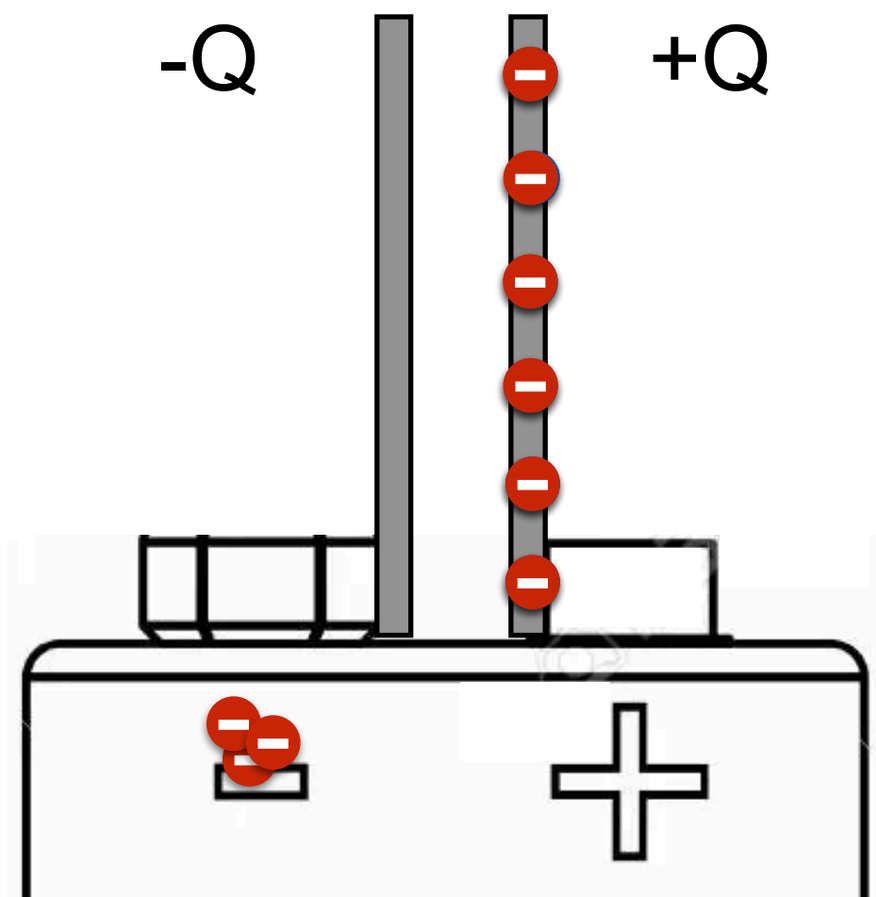
Lecture 12

The Penny Capacitor

Richard made a capacitor by putting two pennies in the gap between the terminals of a 9V battery.

How much charge gets stored on each penny?

Brainstorm : what quantities do we need to know?



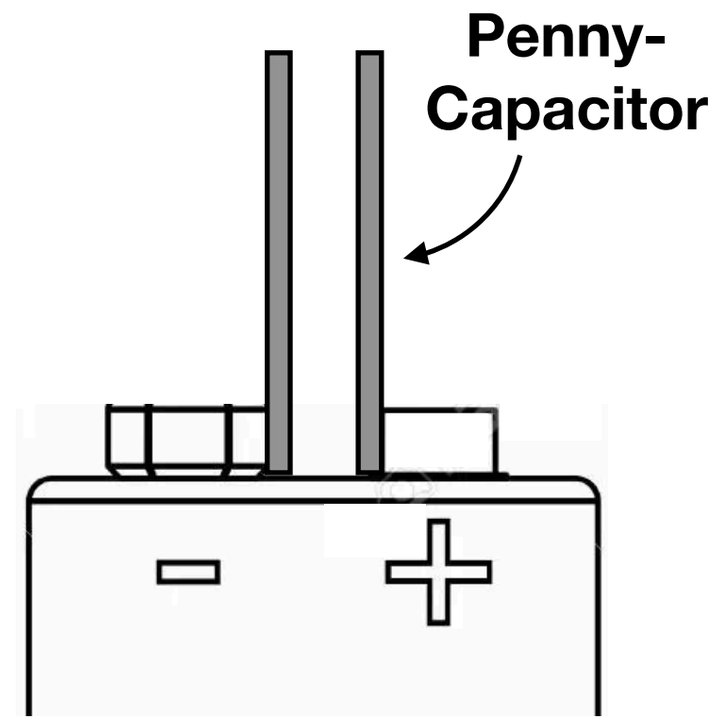
The Penny Capacitor

Capacitance

A penny has a radius of 9.525mm and is 1.52mm thick. It is made of copper.

The inner edges of the terminals of a 9V battery are separated by a distance of 6.55mm.

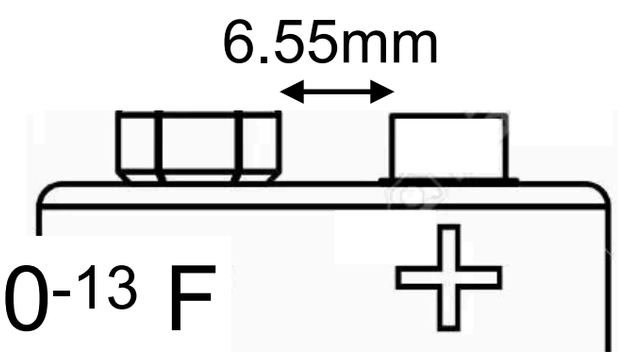
Calculate the capacitance.



$d = 6.55\text{mm} - (2\times1.52\text{mm}) = 3.51\text{mm}$

distance between front faces of the capacitor

$A = \pi\,(0.009525\text{m})^2 = 2.85\times10^{-4}\ \text{m}^2$



$$C = \frac{\varepsilon_0 A}{d} = \frac{8.85\times10^{-12}\,(2.85\times10^{-4})}{3.51\times10^{-3}} = 7.19\times10^{-13}\ \text{F}$$

Calculate the charge.

$$C = \frac{Q}{\Delta V} \rightarrow Q = C\,\Delta V = (7.19\times10^{-13})(9\text{V})$$

$$= 6\times10^{-12}\ \text{C} \qquad \sim 40\text{M}\ e^-$$

Capacitors by DUNDER MIFFLIN, INC

A capacitor is made by covering both sides of a 216mm x 279mm piece of paper with aluminum foil. The paper is a 0.12mm thick.

What is the capacitance of this capacitor?

$$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85\times10^{-12})(0.216\text{m} \times 0.279\text{m})}{0.00012\text{m}} = 4.4 \times10^{-9}\text{F}$$

$$= 4.4 \text{ nF}$$

How much charge will it store when connected to a 9V battery?

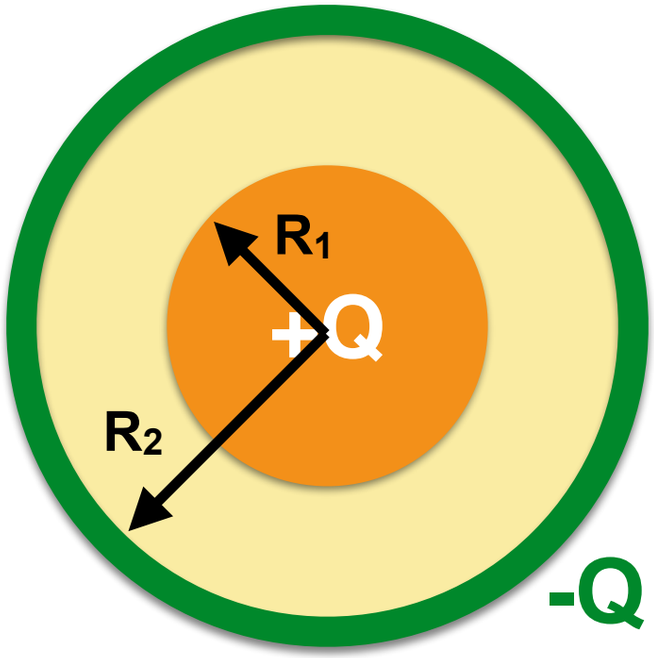
$$Q = C\,\Delta V_c = (4.4 \times10^{-9}\text{F})(9\text{V}) = 4.0 \times10^{-8}\text{C}$$

$$= 40 \text{ nC}$$

Not quite right…

Non-Parallel Plate Capacitors

A capacitor is made from a **spherical ball** (radius R_1) with charge +Q concentric with a **spherical shell** (radius R_2) with charge -Q.



What is the capacitance of this spherical capacitor?

$$C = \frac{Q}{\Delta V}$$

What is the voltage difference between the surfaces?

$$\cancel{V = K\int \frac{dq}{R}}$$

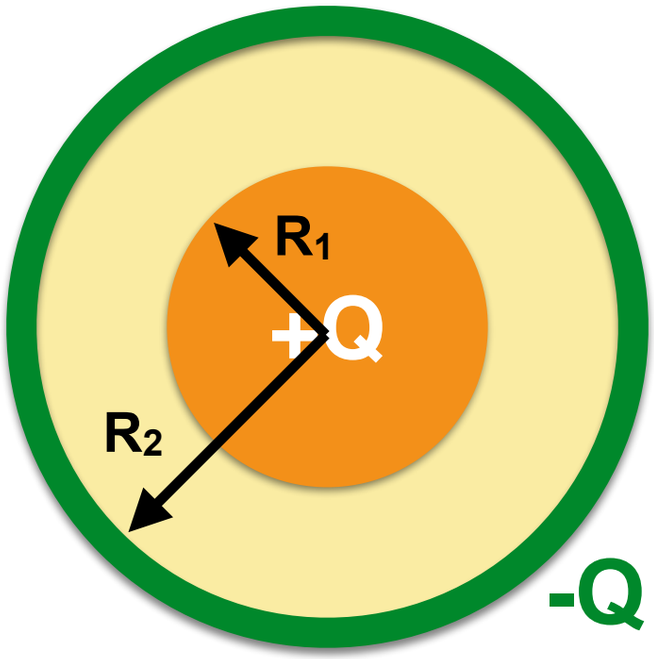
$$\Delta V = -\int_{R_1}^{R_2} \vec{E} \cdot d\vec{s}$$

What is the electric field in the region <u>between</u> the surfaces?

Non-Parallel Plate Capacitors

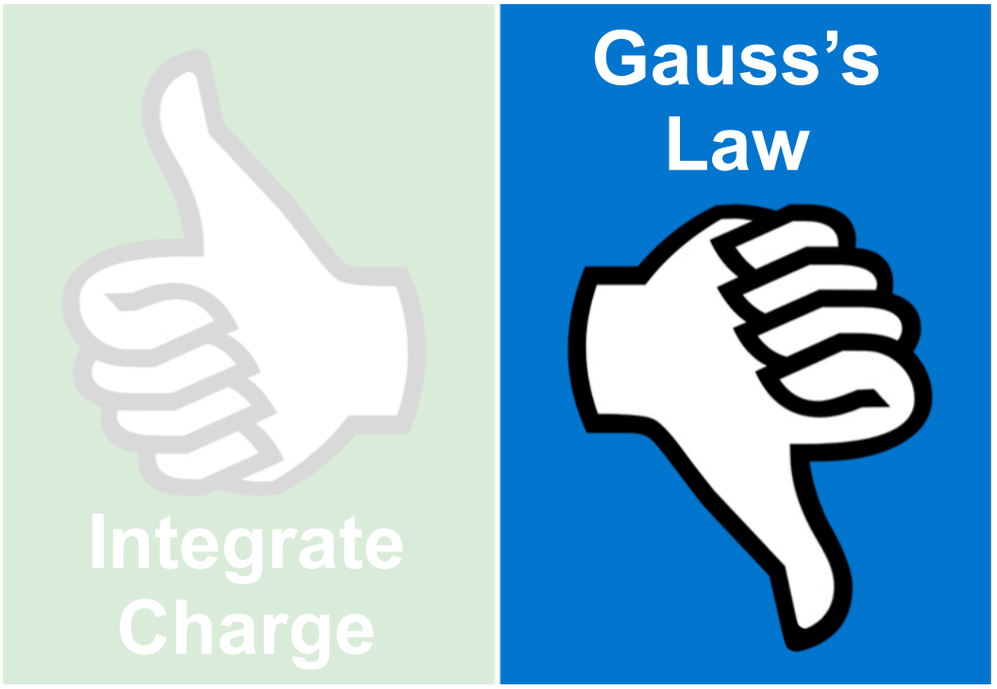
Electric Field Method

A capacitor is made from a **spherical ball** (radius R_1) with charge +Q concentric with a **spherical shell** (radius R_2) with charge -Q.



What is the electric field in the region <u>between</u> the surfaces?

Which method should we try to use?

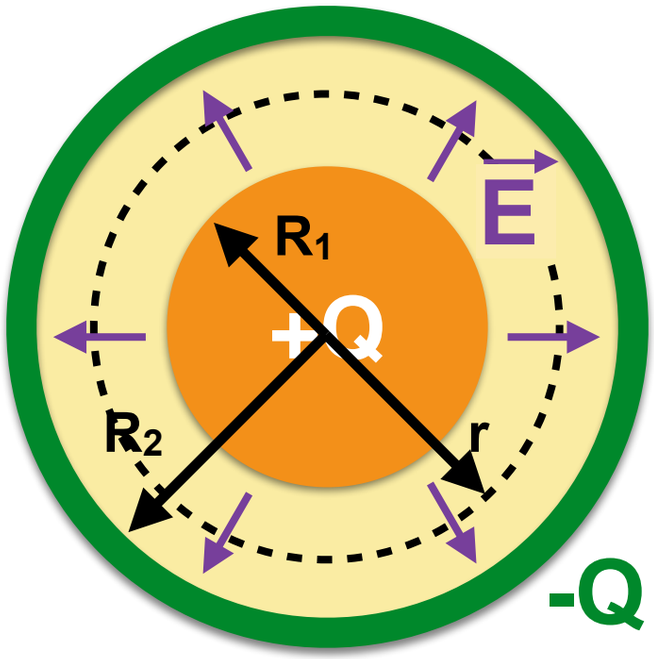


There is *a lot* of symmetry.

Non-Parallel Plate Capacitors

Electric Field

A capacitor is made from a **spherical ball** (radius R_1) with charge +Q concentric with a **spherical shell** (radius R_2) with charge -Q.



What is the electric field in the region <u>between</u> the surfaces?

Derive $\vec{E}$.

$\vec{E}$ is radially outward

Concentric Gaussian Sphere with radius r

Flux : $\oint \vec{E} \cdot dA = \oint E \, dA$

$= E \oint dA = E \, (4 \pi r^2)$

Enclosed charge : +Q

Solution:

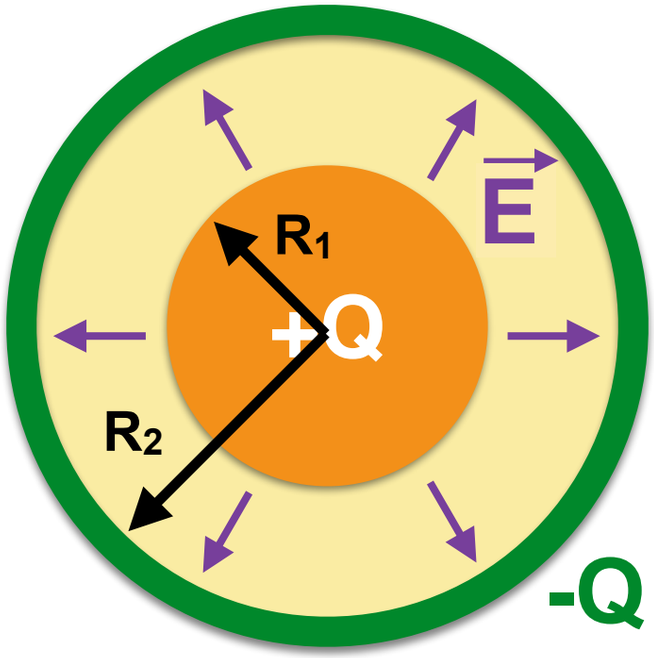
$$\frac{Q}{\varepsilon_0} = E \, (4 \pi r^2)$$

$$\vec{E} = \frac{Q}{4 \pi \varepsilon_0 r^2} \hat{r}$$

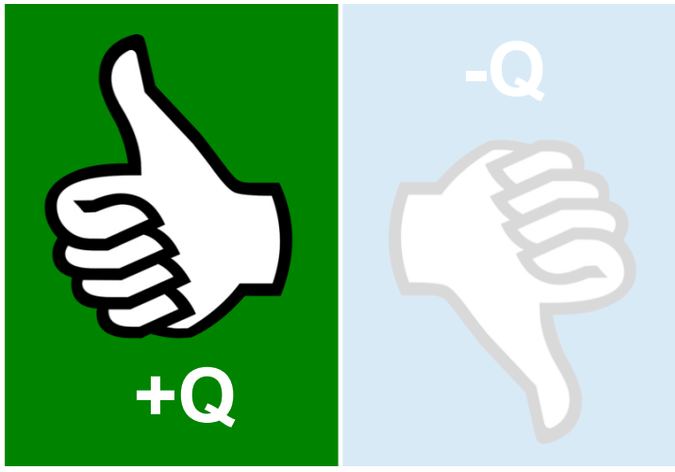
Non-Parallel Plate Capacitors

Potential

A capacitor is made from a **spherical ball** (radius R_1) with charge +Q concentric with a **spherical shell** (radius R_2) with charge -Q.



Which surface is at a higher voltage?



Calculate the voltage difference between the two surfaces.

$$\Delta V = -\int_{R_1}^{R_2} \vec{E} \cdot d\vec{s} = -\int_{R_1}^{R_2} \vec{E} \cdot dr\,\hat{r}$$

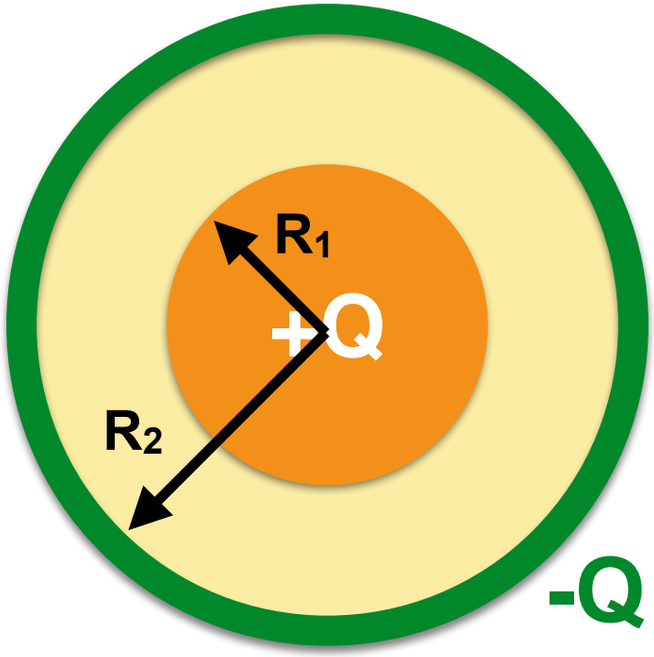
$$\Delta V = -\int_{R_1}^{R_2} \frac{Q\,dr}{4\,\pi\,\varepsilon_0\,r^2}$$

$$\Delta V = \frac{Q}{4\,\pi\,\varepsilon_0}\left(\frac{1}{R_2} - \frac{1}{R_1}\right)$$

Non-Parallel Plate Capacitors

Capacitance

A capacitor is made from a **spherical ball** (radius R_1) with charge +Q concentric with a **spherical shell** (radius R_2) with charge -Q.



What is the capacitance of this spherical capacitor?

$$C = \frac{Q}{\Delta V} = \frac{\cancel{Q}}{\frac{\cancel{Q}}{4\pi\varepsilon_0}\left|\frac{1}{R_2} - \frac{1}{R_1}\right|} = \frac{4\pi\varepsilon_0}{\left|\frac{1}{R_2} - \frac{1}{R_1}\right|}$$

$$= \frac{4\pi\varepsilon_0}{\left|\frac{R_1 - R_2}{R_1 R_2}\right|} = \frac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1}$$

Still only depends on geometry!

Pre-class Slides

Charged Plates, Redux

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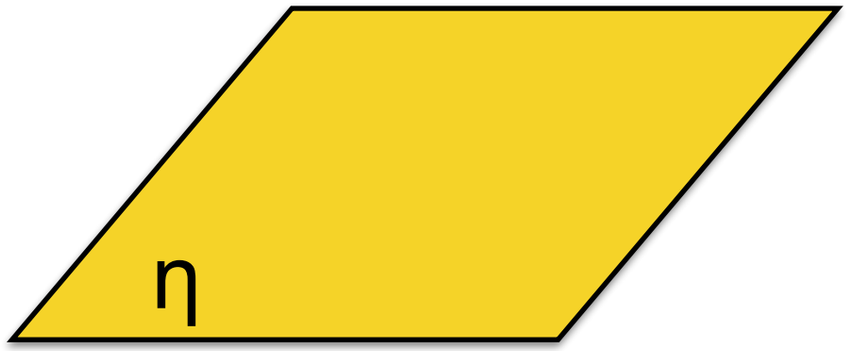
Youtube Playlist

Review : Single Plate Electric Field

A single infinite plate with uniform surface charge density η creates an electric field.



The field has magnitude : $|E| = |\eta| / 2\,\varepsilon_0$

The field points towards the plate if $\eta < 0$

The field points away from the plate if $\eta > 0$

Now Two Plates!

Two "infinitely" large plates are parallel and have opposite charge densities **+η** and **-η**.

The space between the plates is much, much smaller than their size.

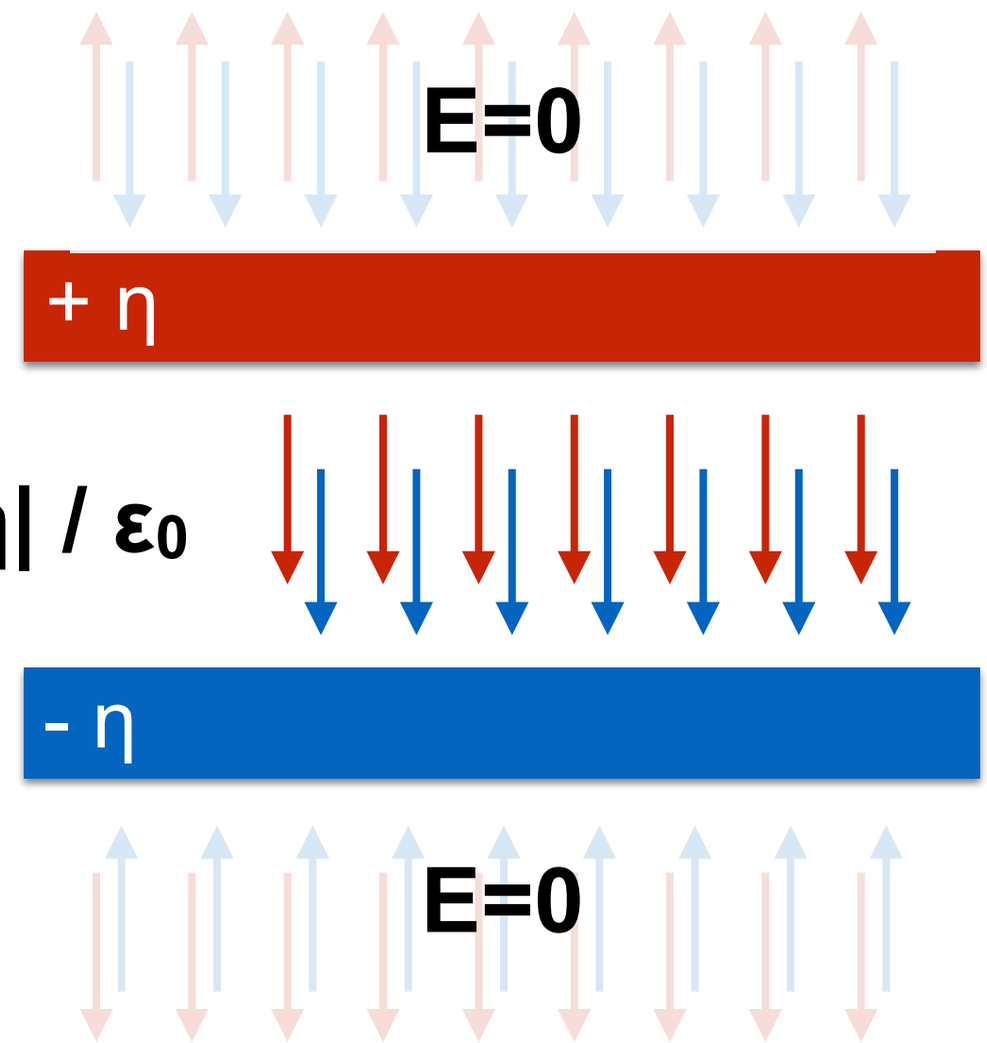
What is the electric field magnitude between the plates?

1. $|\eta| / 4\ \varepsilon_0$
2. $|\eta| / 2\ \varepsilon_0$
3. $|\eta| / \varepsilon_0$
4. $2\ |\eta| / \varepsilon_0$
5. The electric field is zero

Electric field from Two Plates

The magnitude of the field made by each plate is $|\eta| / 2\,\varepsilon_0$

Above the + plate, the two fields cancel each other.



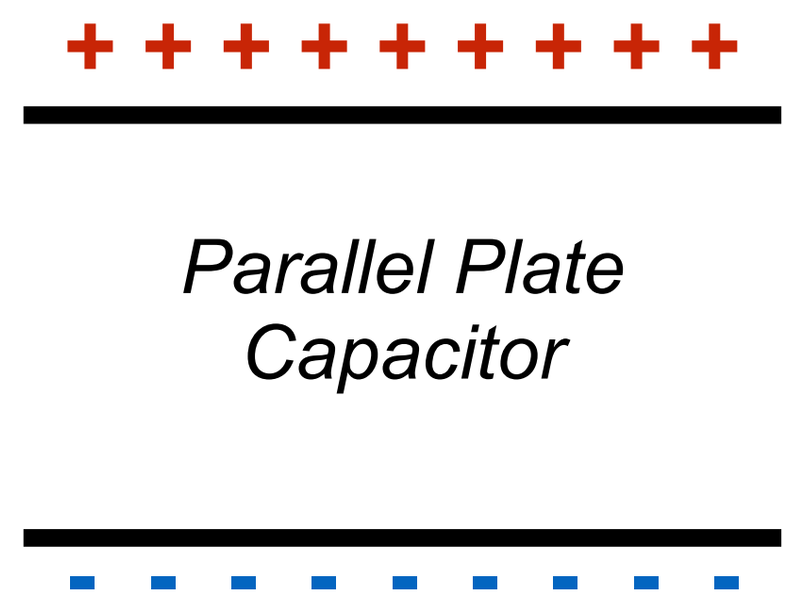
Between the plates, the fields combine!

Below the - plate, the two fields cancel each other.

Parallel Plate Capacitor!

Capacitors

Any object where opposite charges are stored at separate positions to create an electric potential difference between them.



Parallel Plate Capacitor

Useful in circuits for :

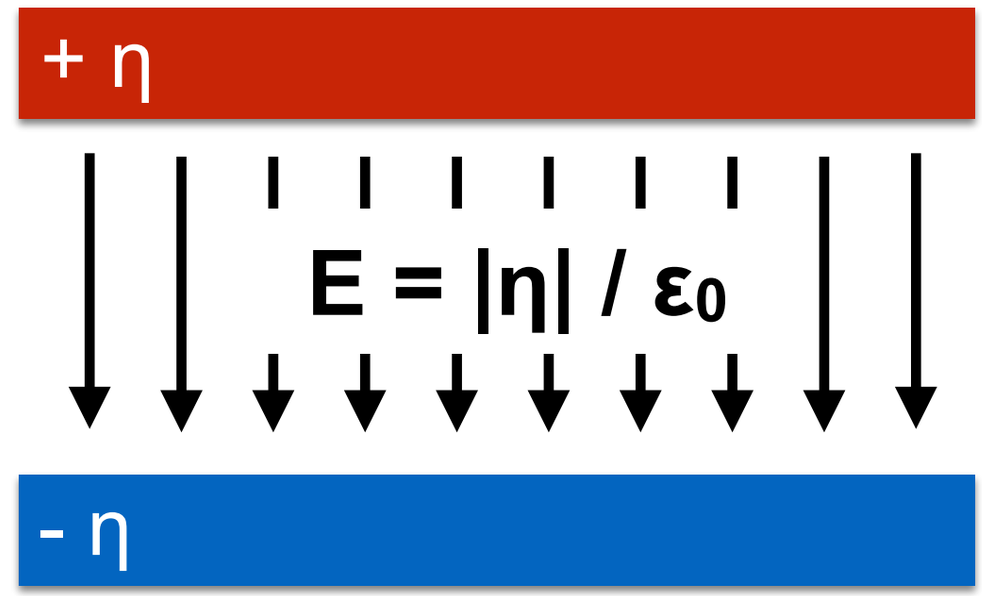
- Storing charge/voltage (short-term)
- Protecting circuits from voltage spikes
- Simple Electronic timing
- Cleaning up (filtering) noisy electrical signals

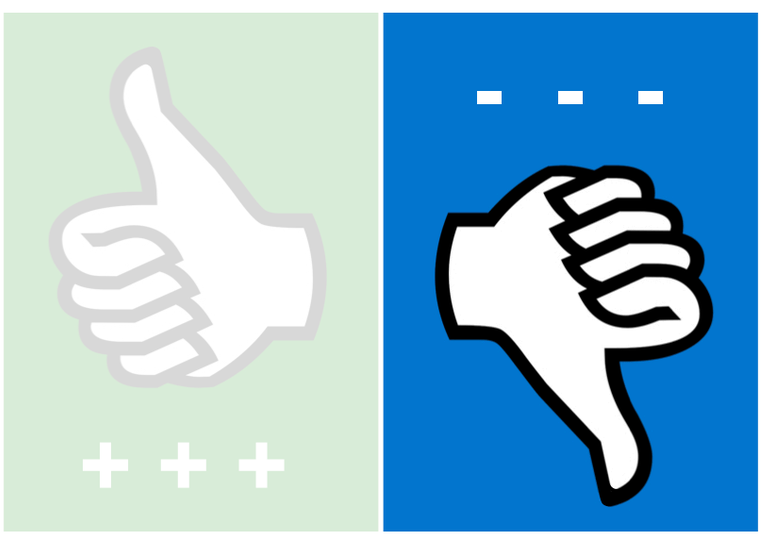
Parallel Plate Capacitor

Potential

The plates of a parallel plate capacitor store opposite charge densities **+η** and **-η**.

The electric potential is lower near which plate?

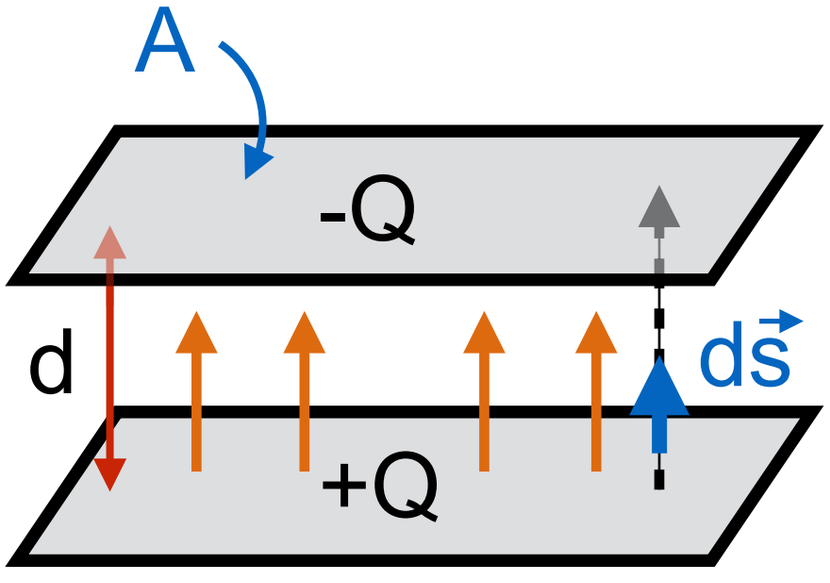




Parallel Plate Capacitor

Voltage Calculation

The plates of a parallel plate capacitor have area A, are separated by distance d, and each holds charge ±Q.



What is the potential difference between the two plates?

$$\vec{E} = \left(\eta / \varepsilon_0\right)\hat{k}$$

$$d\vec{s} = dz\,\hat{k}$$

$$\Delta V = -\int_a^b \vec{E} \bullet d\vec{s} = -\int_0^d (\eta / \varepsilon_0)\,\hat{k} \bullet dz\,\hat{k} = -(\eta / \varepsilon_0)\int_0^d dz$$

$$= -\eta\, d / \varepsilon_0$$

$$= -Q\, d / A\, \varepsilon_0$$

Capacitance

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Capacitance

Charge stored on one capacitor side

$$C = \frac{Q}{\Delta V}$$

Capacitance

Voltage difference across sides of the capacitor

Units : C / V or Farads, F

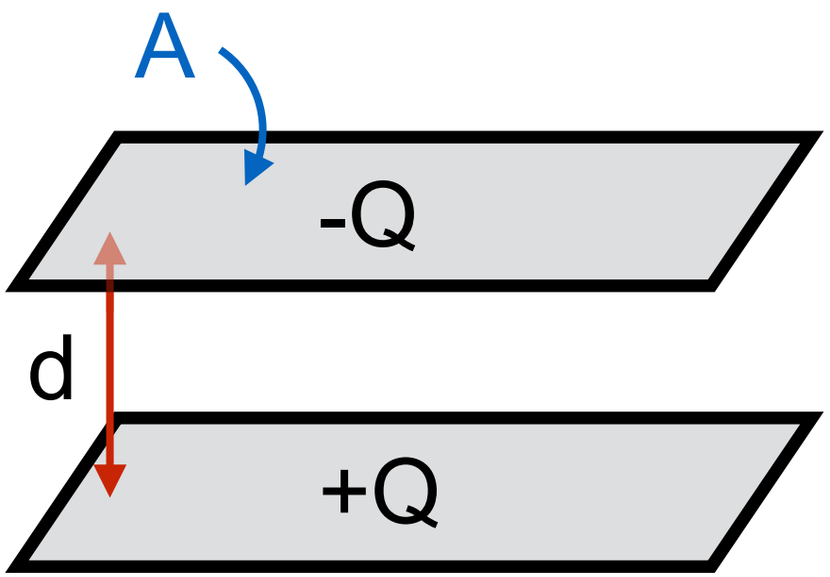
1F is <u>big</u> - most capacitors are μF, nF or pF

10^{-6} 10^{-9} 10^{-12}

Parallel Plate Capacitor

Capacitance Calculation

The plates of a parallel plate capacitor have area A, are separated by distance d, and each holds charge ±Q.



What is the capacitance?

$$C = \frac{Q}{\Delta V} = \frac{Q}{-Q\,d / A\,\varepsilon_0}$$

$$= \frac{-\varepsilon_0 A}{d}$$

Depends on geometry only

Negative?

Came from ΔV calculation.

C just depends on <u>how</u> different the voltage is.

You can drop it!

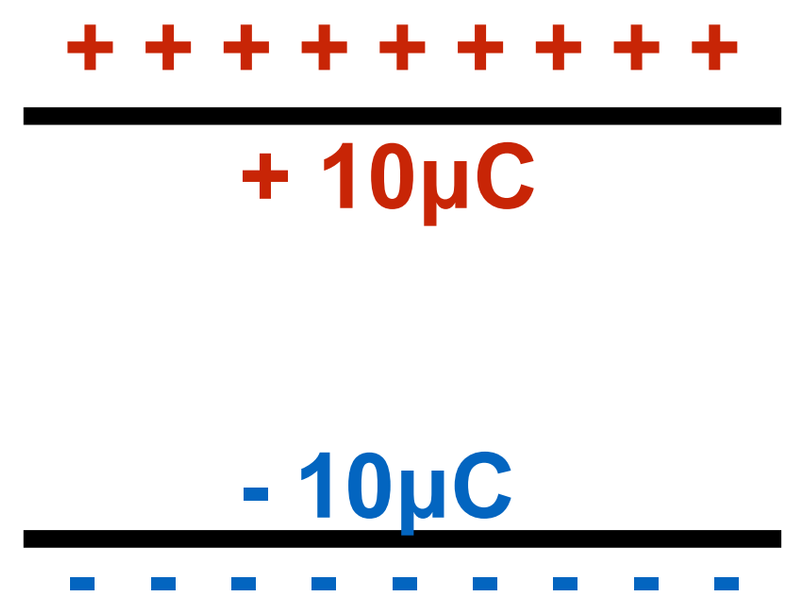
Charge of a Capacitor

The charge stored in a capacitor is 10μC.

What is the net charge of the capacitor?

1. 10 μC
2. 20 μC
3. 5 μC
4. 0 μC
5. -10 μC

0. I don't know



$$C = \frac{Q}{\Delta V}$$

Amount of charge stored on *one* plate